8	**(a)**	photon 'absorbed' by electron	B1	
		photon has energy equal to difference in energy of two energy levels	B1	
		electron de-excites emitting photon (of same energy) in any direction	B1	[3]
	(b) (i)	$E = hc/\lambda$	C1	
		$= (6.63 \times 10^{-34} \times 3 \times 10^{8})/(435 \times 10^{-9})$	C1	
		$= 4.57 \times 10^{-19}$ J (*allow 2 s.f.*)	C1	
		$= (4.57 \times 10^{-19})/(1.6 \times 10^{-19})$ (eV)		
		$= 2.86$ eV (*allow 2 s.f.*)	A1	[4]
	(ii)	arrow pointing in either direction between –3.41 eV and –0.55 eV	B1	[1]